Report R32 — Floquet spectra of the sixteen unitary rules: chaos needs three firing patterns, and the chaotic rules carry exact scars

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-10-14

Abstract

We diagonalise the one-period Floquet operator U of all sixteen unitary rules with the Hadamard gate, for $N = 6 \ldots 14$ sites with open boundaries. For every eigenstate we record the quasienergy, the half-chain entanglement and the gap ratio. All symmetries are resolved first: Krylov sectors, the blocks of the algebra generated by the local gates (which also captures quantum fragmentation and hidden internal symmetries), and the spatial symmetry. The spatial symmetry is the reflection for odd N and, for even N, the half step $\Omega = RL_A$ with $\Omega^2 = U$.

Because U is real orthogonal and a product of two reflections, the chaotic reference is the $\beta = 1$ value $\langle r \rangle = 0.5307$, computed on the half spectrum. We verified this against random reflections.

The sixteen rules sort by the number k of neighbour patterns that fire the gate:

- $k = 3$ (54, 57, 99, 147): the only chaotic rules. $\langle r \rangle = 0.535$ for 57 and 99, and 0.526 for 147, the latter only after symmetry resolution (0.424 without). Each carries an N-independent number of exact eigenstates of every individual gate: two or four product states, and for 54 a single entangled area-law state.
- $k = 1$: exponentially fragmented, with Poisson statistics inside, as expected for tensor products of independent rooms.
- $k = 2$ XOR (150, 105): arithmetic spectra. For 150 the count of distinct quasienergies reproduces R8's 3^m law exactly up to $N = 14$.
- $k = 2$ one-sided: a boundary-driven block tower, with $\langle r \rangle$ near Poisson but drifting upward.
- $k = 0, 4$: trivial.

The literature on these diagnostics for Floquet operators is reviewed, with every reference checked online.

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1 Scope and status

Task: (1) review the literature on eigenstate entanglement, level statistics and scars for Floquet operators; (2) complete the spectral dataset for the 16 unitary rules within 8 hours of compute; (3) per-rule figures; (4) interpretation; (5) a storyline draft and candidate thesis figures, to be shown before any hand-off; (6, optional) the same for the X gate. The classes were to be checked against periodic boundary conditions (user request).

Status at the time of writing. The production runs were stopped at 02:51 by the session's memory reaper, while the first 16384-dimensional sector (rule 51 at $N = 14$) was being processed. They were not restarted without permission.

- **Complete:** obc0, all 16 rules, $N = 6 \ldots 13$; $N = 14$ for 9 of the 16 rules.
- **Missing:** $N = 14$ for the 7 single-sector rules; the $N \geq 15$ stages; all periodic-boundary production runs (only validation runs at $N = 8$ exist); the X gate.
- **The memory fix exists and is validated** (Sec. 7), and a resume plan is given there.

2 The tools, and what they mean for a Floquet operator

Everything in this report is a property of one operator, the one-period evolution $U = L_B L_A$ of the brickwork circuit, and not of a Hamiltonian. This section collects what the literature says the three diagnostics mean in that setting, and where the Hamiltonian intuition has to be dropped.

2.1 Quasienergies, and why there is no "middle of the spectrum"

For a time-periodic Hamiltonian $H(t+T) = H(t)$ or a periodically applied circuit, the stroboscopic dynamics is generated by the Floquet operator $U = \mathcal{T}\exp[-i\int_0^T H(t)\,dt]$, here simply the product of the two gate layers. Its eigenvalues lie on the unit circle, $U\,|\phi_n\rangle = e^{-i\theta_n}\,|\phi_n\rangle$, and the quasienergies θ_n are defined only modulo 2π. There is no ordering, no ground state and no temperature axis.

For a generic interacting driven system the long-time state is the *infinite-temperature* state. D'Alessio and Rigol showed by exact diagonalisation and finite-size scaling that for any non-zero driving period the Floquet operator of a generic lattice system acquires circular-ensemble properties as the system grows. The Floquet Hamiltonian is then non-local and the system heats to infinite temperature [8]. Lazarides, Das and Moessner reached the same conclusion from the structure of Floquet eigenstates, which mix many eigenstates of the undriven Hamiltonian. They also showed that integrable driven systems are instead described by a periodic generalised Gibbs ensemble [9]. Ponte, Chandran, Papić and Abanin found that ergodic systems delocalise in energy space and heat up under both local and global driving [10]. Disorder can prevent this: the Floquet eigenstates of a many-body localised system obey an area law, and a delocalised phase is separated from the MBL phase by a transition seen in the eigenstate entanglement [11].

Consequence for the figures. The Floquet version of the eigenstate thermalisation hypothesis (ETH) [12, 13, 14, 15] says that every eigenstate of a chaotic U looks locally like the infinite-temperature state. Plotted against θ, the eigenstate entanglement of a thermalising circuit is therefore *flat*, a band at the random-state value at every quasienergy. There is no Hamiltonian "dome" that peaks in the middle of the spectrum. A non-thermal (scar-like) eigenstate shows up as an outlier *below* the band, at whatever quasienergy it happens to sit.

2.2 Spectral statistics: circular ensembles and the gap ratio

Dyson's circular ensembles are the random-matrix models for unitary operators [3]: orthogonal (COE, $\beta = 1$), unitary (CUE, $\beta = 2$) and symplectic (CSE, $\beta = 4$). The Bohigas–Giannoni–Schmit conjecture [4] assigns random-matrix level statistics to quantum chaotic systems. The Berry–Tabor conjecture assigns Poisson statistics to integrable ones, with uncorrelated, clustering levels [5]. Which circular ensemble applies is fixed by the antiunitary symmetries *of U itself*. This need not match the class of the instantaneous Hamiltonian: Regnault and Nandkishore showed that a Floquet system can be in the unitary class while its instantaneous Hamiltonian is always orthogonal, and vice versa. They also showed that a drive can thermalise even when the instantaneous Hamiltonian is integrable [6].

The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ of consecutive spacings, introduced by Oganesyan and Huse [2], is independent of the local density of states and so needs no unfolding. Atas *et al.* derived Wigner-like surmises for its distribution

$$P_\beta(r) = \frac{2}{Z_\beta}\,\frac{(r+r^2)^\beta}{(1+r+r^2)^{1+3\beta/2}}, \qquad r \in [0,1], \tag{1}$$

with $Z_1 = 8/27$ and $Z_2 = 4\pi/(81\sqrt{3})$. Their large-matrix values, $\langle r\rangle = 0.5307$ (GOE) and 0.5996 (GUE), are the reference values used here. Poisson statistics give $P_0(r) = 2/(1+r)^2$ and $\langle r\rangle = 2\ln 2 - 1 = 0.3863$ [1]. For a chaotic Floquet operator the quasienergy density is uniform, so the ratio is doubly convenient.

Two caveats from the literature are central to this report:

- **Symmetry sectors must be resolved.** Superposing independent spectra from different symmetry sectors removes level repulsion between them and pulls $\langle r\rangle$ towards the Poisson value, even if every sector is chaotic. Giraud, Macé, Vernier and Alet derived the gap-ratio surmises for such superpositions and used them to detect hidden symmetries,

including in periodically driven spin chains [7]. ETH, too, is a statement per symmetry sector. Mondaini *et al.* made exactly this point in their comment [21] on a claimed ETH counterexample [20].

- **In fragmented systems the Krylov sectors play the role of symmetry sectors.** Hilbert-space fragmentation (HSF) splits the space into exponentially many invariant subspaces that are not labelled by conventional symmetries [26, 27]. In the commutant language of Moudgalya and Motrunich [28], the invariant subspaces are the blocks of the algebra generated by the local terms (or gates). Its commutant has exponential dimension, and the product-basis Krylov sectors ("classical" fragmentation) can be refined further by quantum fragmentation [28]. Level statistics are meaningful only inside one such block. Pooling blocks measures their number, not their chaoticity.

2.3 Eigenstate entanglement

For a random pure state on $\mathbb{C}^m \otimes \mathbb{C}^n$ with $m \leq n$, Page's mean entanglement entropy is $S_{\text{Page}} \approx \ln m - m/(2n)$, short of the maximum by about half a unit of information [16]. Eigenstates of chaotic Hamiltonians approach this value in the middle of the spectrum. Conservation laws lower it: Vidmar and Rigol showed that for random states at fixed particle number away from half filling, the average entropy falls short of the maximum by an amount growing as the square root of the volume [17]. For a fragmented system the right benchmark is therefore a random state *inside the relevant block*, not S_{Page} of the full space.

Volume-law eigenstates signal thermalisation. Area-law or otherwise anomalously weakly entangled eigenstates embedded in a thermal spectrum are the defining signature of quantum many-body scars (QMBS). QMBS were identified as the origin of the persistent revivals in the Rydberg-atom experiment [18], via scarred eigenstates of the PXP model [19]. Embedding constructions produce them systematically [20]. Reviews cover the semiclassical picture [22], exact results [23], and the quasiparticle perspective [24]. Scars also exist intrinsically in driven systems: Sugiura, Kuwahara and Saito proved that a PXP-type Floquet model hosts exact scar eigenstates of U at specific driving periods, a rigorous breakdown of Floquet ETH [25].

Degeneracies make single-eigenvector entanglement ill-defined. If an eigenvalue of U is g-fold degenerate, any orthonormal basis of the eigenspace is equally valid, and the entanglement of "the" eigenvectors depends on that arbitrary choice. Earlier R8 found that 43–74 % of the stored rule-150 eigenstates lie in such eigenspaces, with multiplicity up to 35. This report therefore (i) resolves all symmetries first, so the remaining basis freedom is as small as possible, (ii) records g for every state, and (iii) adds a basis-independent quantity for every eigenspace: the Haar-averaged Rényi-2 entropy (Sec. 3).

2.4 Quantum circuits and automata in particular

Brickwork Floquet circuits are the setting in which many-body quantum chaos has been connected analytically to random-matrix theory. Kos, Ljubotina and Prosen derived RMT spectral correlations for periodically kicked many-body systems [29]. Chan, De Luca and Chalker solved a Floquet circuit of random two-site gates (depth two, like ours) in the large-q limit, including its spectral form factor and entanglement growth [30]. The circuits studied here are the opposite extreme: *one fixed* controlled gate, repeated.

Automaton-based circuits are a known source of non-ergodic eigenstates. Gopalakrishnan and Zakirov constructed facilitated quantum cellular automata, mostly from Clifford gates. Their eigenstate entanglement differs from generic chaotic systems in all five models, and some models have exponentially many eigenstates with disentangled sites, tied to reflection symmetry [31]. Iadecola and Vijay showed that deforming a classical reversible cellular automaton gives generically non-integrable Floquet systems that inherit exact non-thermal eigenstates from the

automaton's periodic orbits [32]. The automata themselves (our $V = X$ circuits) include the interacting integrable Rule 54 [33] and the Floquet-PXP Rule 201, whose solitons are domain walls between three vacua [34].

2.5 Two structural facts used throughout

U is real orthogonal and a product of two reflections. Both gates, H and X, are real symmetric involutions, so every controlled gate is one and each layer $L_{A,B}$ is a real symmetric involution (a reflection). Hence $U = L_B L_A$ is real orthogonal. Two consequences:

1. Complex conjugation K commutes with U, so within every symmetry-resolved block the spectrum is symmetric under $\theta \to -\theta$. The independent spectral data are the $|\theta| \in [0, \pi]$. Ratios computed over the full circle would count each spacing twice and add spurious small gaps where mirror pairs approach 0 and π.

2. By the two-subspace theorem [35, 36], the space decomposes into the intersections of the ± 1 eigenspaces of L_A and L_B (eigenvalues ± 1 of U) and two-dimensional subspaces on which U rotates by $\pm 2\varphi_j$, where the φ_j are the principal angles between the $+1$ eigenspaces of the two layers. For generic reflections these angles follow a real ($\beta = 1$) Jacobi ensemble, whose bulk statistics are those of the GOE. We verified the reference value directly with the same routine used for the data: products of two Haar-random reflections of size 300–800 give $\langle r \rangle = 0.533 \pm 0.004$ on the half spectrum, compared with 0.535 ± 0.005 for a GOE control and 0.604 ± 0.005 for a GUE control.

In addition $T = L_A K$ is an antiunitary with $TUT^{-1} = U^{-1}$ and $T^2 = 1$, a time-reversal symmetry, and also points to the orthogonal class. The chaotic reference throughout is therefore $\langle r \rangle_{\mathrm{GOE}} = 0.5307$, not the CUE value.

The X gate makes U a permutation. With $V = X$ every basis state is mapped to a single basis state. On a cycle of length ℓ the spectrum is the ℓ-th roots of unity, a picket fence with $r \equiv 1$, and the eigenvectors are Fourier modes along the cycle. Each Krylov sector of a permutation is a single cycle, so this holds sector by sector. The commutant study R-T26 found exactly this, $P(r) = \delta(r - 1)$, for rule 156. Neither Poisson nor RMT applies, and an X-gate $\langle r \rangle$ measures orbit arithmetic, not chaos.

3 Methods

3.1 Model and conventions

The sixteen unitary rules are the tuples $(r_{00}, r_{01}, r_{10}, r_{11}) \in \{I, V\}^4$, Wolfram numbers 51, 54, 57, 60, 99, 102, 105, 108, 147, 150, 153, 156, 195, 198, 201 and 204, with $V = H$ throughout this report (Sec. 6 for $V = X$). Boundary conditions are open with fixed vacuum neighbours (obc0). One period is $U = L_B L_A$, with L_A the gates on the odd (1-based) sites, applied first, and L_B those on the even sites. The quasienergy convention is $U|\phi\rangle = e^{-i\theta}|\phi\rangle$, $\theta \in (-\pi, \pi]$. The bipartition is $A =$ sites $1 \ldots \lfloor N/2 \rfloor$.

3.2 Pipeline and independent cross-checks

The diagonalisation is done in Julia by `HSF/generate_spectral_data.jl`, an optimised successor of the earlier `generate_EE_data.jl`, which is left untouched. The analysis is done in Python, in `code/qca_fragmentation/spectral/`.

- **Two independent constructions of U** agree. The Julia operator is built from Kronecker products of local gates. The Python operator comes from this repository's exact $\mathbb{Z}[1/\sqrt{2}]$ engine (`core/cycle.py`). Their quasienergy multisets agree to $< 10^{-14}$ for all 16 rules at $N = 8$ (test `test_engine_matches_julia_spectrum`, $N = 6, 7, 8$). Comparisons of angle multisets cut the circle in the middle of the largest gap. Sorting raw angles is unsafe because a level at $\theta = 0$ computed as -10^{-16} wraps to 2π. A first version of the test fell into exactly this trap and reported a spurious disagreement.

- **Against the stored HSF eigendata** (complex `eigen` per sector): quasienergies agree to 10^{-14}. The entropies agree to 10^{-15} for every globally non-degenerate level, the only case where the old entropies are basis independent.

3.3 Diagonalisation

U is real orthogonal (Sec. 2). Each block is diagonalised by

1. solving the real symmetric problem $C = (A + A^T)/2$, whose eigenvalues are $\cos\theta$, with LAPACK's MRRR solver ($O(n)$ workspace);

2. grouping the $\cos\theta$ into clusters (tolerance 10^{-9}) and diagonalising the antisymmetric part $(A - A^T)/2$ inside each cluster, which separates θ from $-\theta$.

This yields exactly orthonormal eigenvectors, also inside degenerate eigenspaces, which a general complex eigensolver does not guarantee. It is also several times faster. The three dense products the cluster loop needs are done as single matrix–matrix multiplications rather than per cluster (that step was memory-bound and took half the run time). On the 12th-gen i7 used here, dense LAPACK is fastest with 4–6 BLAS threads and degrades sharply at 20. Every run stores the largest eigen-residual $\max |Uv - e^{-i\theta}v|$ and the orthonormality error.

3.4 Symmetry resolution

Each eigenvector carries three labels, all found and verified at run time:

1. **Krylov sector**: connected component of the support of U (classical fragmentation).

2. **Gate-algebra block**: the decomposition of each sector by the commutant of the bond algebra $\mathcal{A}_0 = \langle g_1, \ldots, g_N \rangle$ generated by the local gates [28]. This includes quantum fragmentation and hidden internal symmetries. It is found without prior knowledge of the symmetry:

   (a) diagonalise a generic element $K = \sum_i a_i g_i$ (random $a_i \in [1, 2]$, fixed seed);

   (b) transform a second generic element $K' = \sum_i b_i g_i$ into that eigenbasis;

   (c) take the connected components of the transformed K' as the blocks.

   If the commutant is Abelian, each eigenvector of K lies in one block, and K' connects everything within a block. Isomorphic blocks (non-Abelian multiplicity) are merged, which is harmless: they give exactly degenerate copies of one spectrum, removed in the statistics. *Check*: the largest matrix element of $Q^T U Q$ between different blocks is stored (`max_offblock`).

3. **Spatial label.**

   - For odd N and mirror-symmetric rules ($r_{01} = r_{10}$), the reflection R commutes with U, and every eigenvector gets a parity ± 1.

- For even N, R exchanges the two layers ($RL_AR = L_B$), so R alone is no symmetry. The half step $\Omega = RL_A$ is one, however: it commutes with U and $\Omega^2 = U$ (both asserted at run time). Inside an eigenspace of U, $\Omega e^{i\theta/2}$ is Hermitian with eigenvalues $s = \pm 1$, and s is the label.

Labels are obtained by diagonalising the symmetry inside each degenerate eigenspace of U, so the stored eigenvectors are symmetry adapted. The symmetry is used for a block only if the block is checked to be invariant under it. The invariance residual is stored.

Time reversal ($T = L_AK$, $T^2 = 1$) fixes the reference ensemble to $\beta = 1$ (Sec. 2).

3.5 Statistics

Gap ratios are computed per fully resolved block on the de-duplicated half spectrum $|\theta| \in [0, \pi]$. Only blocks with at least 20 distinct half-levels are used, and the ratios of all such blocks are pooled. For comparison the same statistic is computed at two coarser resolutions: per Krylov sector only, and for the whole spectrum as one list.

3.6 Entanglement and the fair reference

For every eigenvector the half-chain von Neumann entropy S_{vN} is stored. For every degenerate eigenspace of dimension g (inside its block and label) the basis-independent Haar-averaged Rényi-2 entropy is stored as well,

$$S_2^{\mathrm{Haar}} = -\ln \frac{\operatorname{Tr} P_A^2 + \operatorname{Tr} P_B^2}{g(g+1)}, \qquad P_A = \operatorname{Tr}_B P,\ P_B = \operatorname{Tr}_A P, \tag{2}$$

where P is the projector onto the eigenspace. This follows from $\mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}] = P^{\otimes 2}(1 + \mathrm{SWAP})/(g(g+1))$, and for $g = 1$ it is the ordinary Rényi-2 entropy of the eigenvector. The same formula with P the projector onto the whole *block* gives $S_2^{\mathrm{rand}}(\mathrm{block})$, the entanglement of a random state inside that block. That is the fair reference for a fragmented system (Sec. 2), and the ratio

$$\eta = S_2^{\mathrm{Haar}} / S_2^{\mathrm{rand}}(\mathrm{block}) \tag{3}$$

is ≈ 1 for a thermal eigenstate and $\ll 1$ for an atypically weakly entangled, scar-like one. η is defined for blocks of dimension ≥ 2. States with $\eta < 0.5$ in blocks of dimension ≥ 20 are counted as scar-like.

4 Results: the sixteen rules with the Hadamard gate

4.1 The organising principle: how many neighbour patterns fire the gate

Count, for each rule, the number k of the four neighbour patterns $(x_{i-1}, x_{i+1}) \in \{00, 01, 10, 11\}$ on which the Hadamard acts. For $k = 2$, also record whether the firing set depends on only one neighbour ("one-sided": $60, 102, 153, 195$) or on both ("XOR": 150 fires on $01, 10$, and 105 on $00, 11$). This single combinatorial number sorts the 16 rules into classes with distinct spectral signatures (Table 1, Fig. 1):

k	W	N	sym.	sectors	blocks	$\langle r\rangle_{\text{none}}$	$\langle r\rangle_{\text{full}}$	$\#r$	distinct $\theta/2^N$	$\bar{\eta}_{\text{big}}$	singlets
0	204	14	R	16384	16384	–	–	0	6.1e-05	–	0
1	108	14	Omega	3329	5710	0.399	0.383	1106	0.0659	0.61	2074
1	156	14	none	987	5740	0.368	–	0	0.0222	0.00	2439
1	198	14	none	987	5740	0.368	–	0	0.0222	0.00	2439
1	201	14	Omega	1897	4161	0.391	0.395	2120	0.137	0.57	1765
2 one-sided	60	14	none	15	105	0.384	0.395	7732	0.492	0.68	24
2 one-sided	102	14	none	15	105	0.384	0.395	7732	0.492	0.68	24
2 one-sided	153	13	none	1	14	0.385	0.414	3998	0.985	0.77	2
2 one-sided	195	13	none	1	14	0.385	0.414	3998	0.985	0.75	2
2 XOR	105	14	Omega	8	22	0.374	0.357	3616	0.133	0.65	7
2 XOR	150	14	Omega	8	22	0.441	0.420	3616	0.133	0.63	7
3	54	13	R	2	4	0.422	0.510	4030	0.985	0.78	1
3	57	13	none	1	3	0.535	0.535	4094	1	0.95	2
3	99	13	none	1	3	0.535	0.535	4094	1	0.95	2
3	147	13	R	1	6	0.424	0.526	4022	0.982	0.84	4
4	51	13	R	1	8192	–	–	0	0.000244	–	8192

Table 1: Per-rule summary at each rule's largest computed N (obc0, gate H), generated by `spectral.analysis.write_table`. sym.: spatial symmetry resolved (R for odd N; the half-step Ω for even N). sectors/blocks: Krylov sectors, and fully resolved gate-algebra blocks. $\langle r\rangle_{\text{none}}$: whole spectrum as one list. $\langle r\rangle_{\text{full}}$: pooled over fully resolved blocks with ≥ 20 distinct half-levels, from $\#r$ ratios. distinct $\theta/2^N$: number of distinct quasienergies over the whole spectrum. $\bar{\eta}_{\text{big}}$: mean S_2/S_2^{rand}(block) in the largest block. singlets: one-dimensional gate-algebra blocks inside Krylov sectors of dimension ≥ 3, i.e. exact eigenstates of every individual gate.

class	rules	spectral signature (obc0)
$k = 0$	204	$U = I$ (identity)
$k = 1$	108, 156, 198, 201	exponential fragmentation; Poisson inside the pooled blocks
$k = 2$ one-sided	60, 102, 153, 195	a tower of gate-algebra blocks; $\langle r\rangle$ near Poisson, drifting up with N
$k = 2$ XOR	105, 150	arithmetic spectrum, 3^m distinct quasienergies (W150)
$k = 3$	54, 57, 99, 147	**chaotic (GOE)**, plus a fixed number of exact gate-singlet eigenstates
$k = 4$	51	$U = H^{\otimes N}$, non-interacting

4.2 $k = 3$: chaos, with exact scars that do not grow

The four rules on which the gate fires on all but one pattern are the only chaotic ones.

- **57 and 99** (mirror images; computed independently, identical to all digits): $\langle r\rangle = 0.535$ at $N = 13$, on the GOE value 0.5307 from $N = 10$ on: $0.525, 0.523, 0.527, 0.535$ for $N = 10 \ldots 13$. $P(r)$ follows the $\beta = 1$ surmise over the whole range (Fig. 2). The eigenstates in the big block are thermal *within that block*: $\bar{\eta} = 0.95$.
- **147**: $\langle r\rangle_{\text{full}} = 0.526$ ($N = 13$), but only after resolving the reflection (odd N) or the half step Ω (even N). Unresolved, the same spectrum gives 0.424. This is the textbook symmetry-mixing effect, and here it is the difference between "integrable" and "chaotic".
- **54**: $\langle r\rangle_{\text{full}}$ rises steadily, $0.441, 0.456, 0.502, 0.510$ for $N = 10 \ldots 13$, towards GOE. Its frozen vacuum $|0 \ldots 0\rangle$ is a separate Krylov sector.

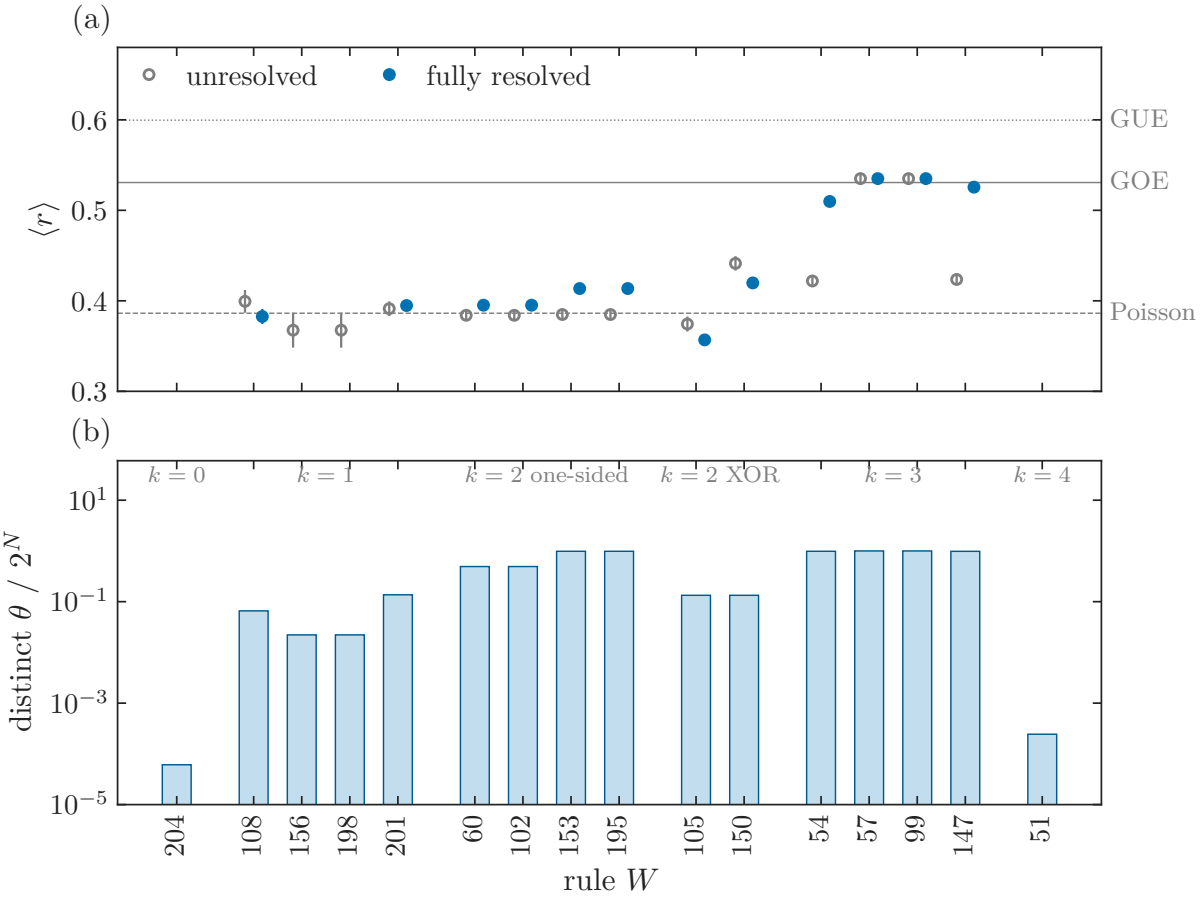


Figure 1: (a) Gap ratio of every rule at its largest N: unresolved (whole spectrum, open circles) and fully resolved (Sec. 3.4, filled). Lines: Poisson, GOE ($\beta = 1$, the class of U), GUE. (b) Fraction of distinct quasienergies in the whole spectrum (log scale). Rules grouped by the number k of firing neighbour patterns.

Gate singlets. Inside the chaotic block, each of these rules carries one-dimensional blocks of the gate algebra: states that are eigenvectors of *every individual gate* g_i, hence of U for any order and any number of gate applications. Their number does not depend on N ($N = 6 \ldots 13$):

- two for 57 and 99, and four for 147, all *product* states ($S = 0$) at $\theta \in \{0, \pi\}$;
- exactly one for 54, at $\theta = 0$, which is *entangled*, with $S_{\mathrm{vN}} = 0.29$ at every N: an area-law state.

These are exact many-body scars in the sense of Refs. [20, 23]: non-thermal eigenstates embedded in a GOE spectrum. The embedding is exact here, by the gate algebra, not approximate. Because they are eigenstates of every gate, they are also stationary under any reordering of the circuit, which a Hamiltonian scar cannot offer.

N-stable outliers in 57/99. Besides the singlets, 57 has a handful of anomalously weakly entangled eigenstates in the big block ($\eta \approx 0.2$): 4 at $N = 11$ and 13, and 10 at $N = 12$. Their quasienergies are stable with N ($\theta/\pi = \pm 0.6877 \to \pm 0.6879$ from $N = 11$ to 13, and ± 0.3122 at $N = 12$ and 13). They do not form an equally spaced tower, and their number does not grow. This points to a localised structure (a single defect in a frozen background) rather than a PXP-type scar band. In 147, by contrast, the low-η states are numerous (72–83) and spread over θ: the finite-size tail of the thermal distribution.

4.3 $k = 1$: exponential fragmentation, Poisson inside

Rules 108, 156, 198 and 201 fire on a single pattern and have exponentially many Krylov sectors (R2, R19). With the gate algebra resolved, the sectors split further: at $N = 14$, 156 has 987 sectors and 5740 blocks, and 201 has 1897 sectors and 4161 blocks. Most blocks are too small for level statistics. 156 and 198 never reach 20 distinct half-levels in one block up to $N = 14$. Where blocks are large enough (108, 201), $\langle r \rangle$ pooled over them is Poissonian and N-independent: $0.384, 0.385, 0.386, 0.383$ for 108 at $N = 11 \ldots 14$. This matches the structure found in the commutant study (R-T18/R-T27): a block is a tensor product of independent rooms, so its quasienergies are sums of room quasienergies, and sums of independent spectra are Poissonian. The gate singlets here are exponentially many (2074 for 108 and 2439 for 156 at $N = 14$, roughly doubling with each added site). These are the dark states of the rooms.

4.4 $k = 2$, XOR: arithmetic spectra

150 and 105 have only a handful of Krylov sectors (the domain-wall shells, R8) and large blocks, but their spectra are *arithmetic*. The number of distinct quasienergies of 150 is exactly 3^m for even N and $(3^m + (-1)^m)/2$ for odd N ($m = \lceil N/2 \rceil$) at every $N = 6 \ldots 14$, reproducing R8's law from an independent diagonalisation. 105 shares the even-N counts but has far fewer at odd N (251 against 1093 at $N = 13$). Up to 95 % of the eigenstates lie in degenerate eigenspaces, and spacings repeat exactly. $\langle r \rangle$ (0.42 and 0.36) is therefore no measure of chaos here. The right description is the free wall walk with sign defects of R8. Each Krylov sector contains one gate singlet at $\theta = 0$, an entangled dark state (S between 0.13 and 1.14).

4.5 $k = 2$, one-sided: a boundary-driven tower, statistics undecided

In 60/102 (gate fires iff the left or right neighbour is 1) and 153/195 (iff it is 0), the gate on a site is controlled by one neighbour only. With open boundaries the gate algebra then produces a tower:

- 153/195: $N + 1$ blocks of sizes $2^{N-1}, 2^{N-2}, \ldots, 2, 1$, plus two singlets;
- 60/102: $2N - 4$ singlets and many small blocks.

At $N = 5$ the dominant basis states of block j (153) are those whose first site from the left in the -1 eigenstate of H is site j. The sites to its left are frozen in the dressed $+1$ state, and the right part is a smaller copy of the chain. The entanglement accordingly falls into bands, one per block (Fig. 4). Within the blocks, $\langle r \rangle$ is near Poisson but *increases* steadily with N: for 60, $0.334, 0.351, 0.360, 0.369, 0.381, 0.395$ for $N = 9 \ldots 14$; for 153, 0.37–0.414 for $N = 9 \ldots 13$. Integrability and a slow crossover towards GOE cannot be told apart at these sizes. The tower itself is a boundary effect: on a ring it is absent (Sec. 5).

4.6 $k = 0$ and $k = 4$: trivial

204 is the identity (one distinct quasienergy). In 51 every site gets H in its layer, so $U = H^{\otimes N}$ exactly. Its spectrum is $\{0, \pi\}$, every state is a gate singlet, and the eigenstates are products of single-site H eigenstates, with $S = 0$.

5 Periodic boundary conditions

The validated pbc pipeline (`generate_spectral_data_v2.jl`, even N only, with translation by two sites and the site-centred reflection resolved) has so far been run only for validation at $N = 8$. There, its spectra agree with the exact Python engine to 10^{-14}. These runs already show the decisive structural point: on a ring the one-sided rules lose their block tower (195: 2

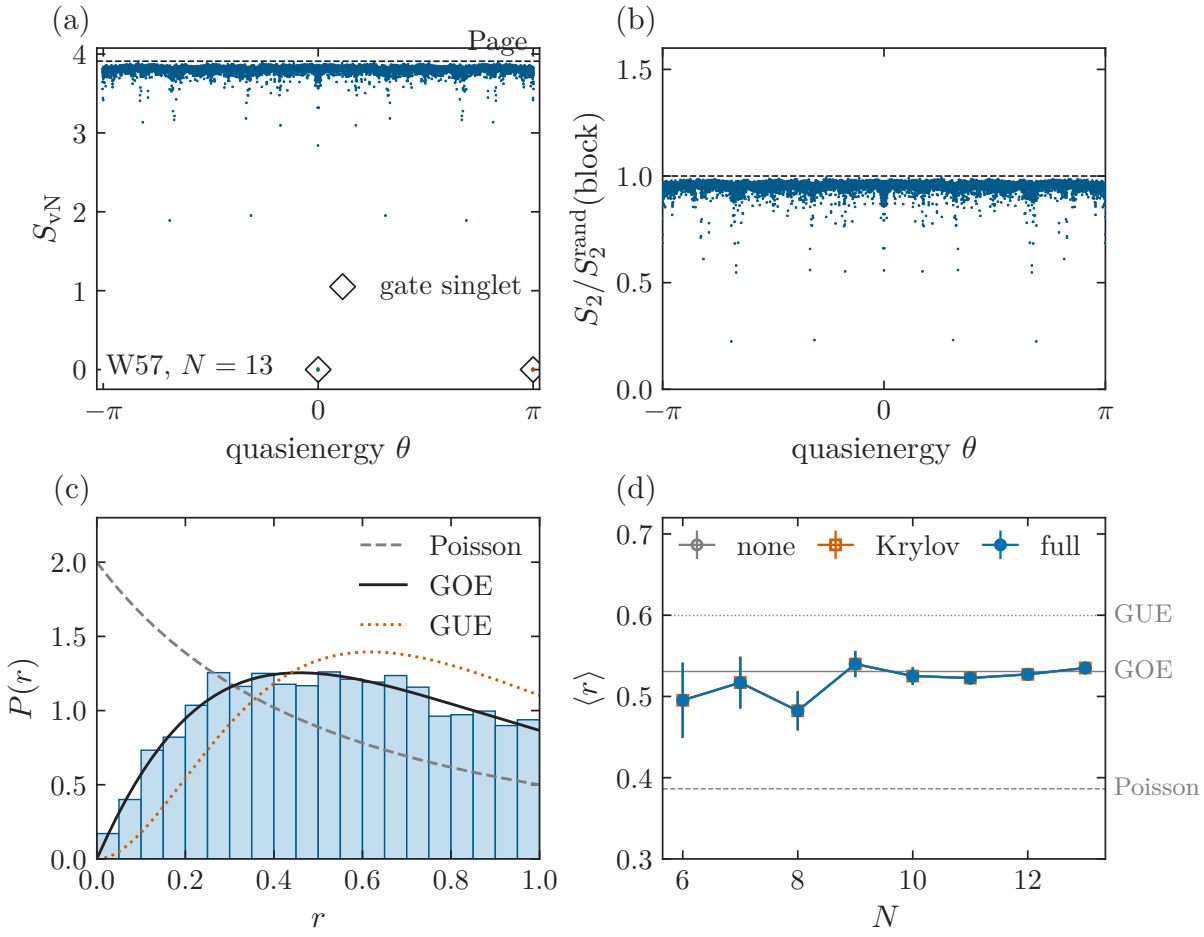


Figure 2: Rule 57 ($k = 3$), $N = 13$, obc0. (a) Half-chain entanglement against quasienergy. Coloured: non-degenerate states, by block. Grey: states in a degenerate eigenspace, whose entanglement depends on the basis. Diamonds: gate singlets (exact eigenstates of every gate; product states at $\theta = 0, \pi$). (b) $\eta = S_2^{\mathrm{Haar}}/S_2^{\mathrm{rand}}$(block), basis independent. (c) $P(r)$ of the fully resolved half spectrum against the Poisson, GOE and GUE surmises. (d) $\langle r \rangle$ against N at three resolution levels.

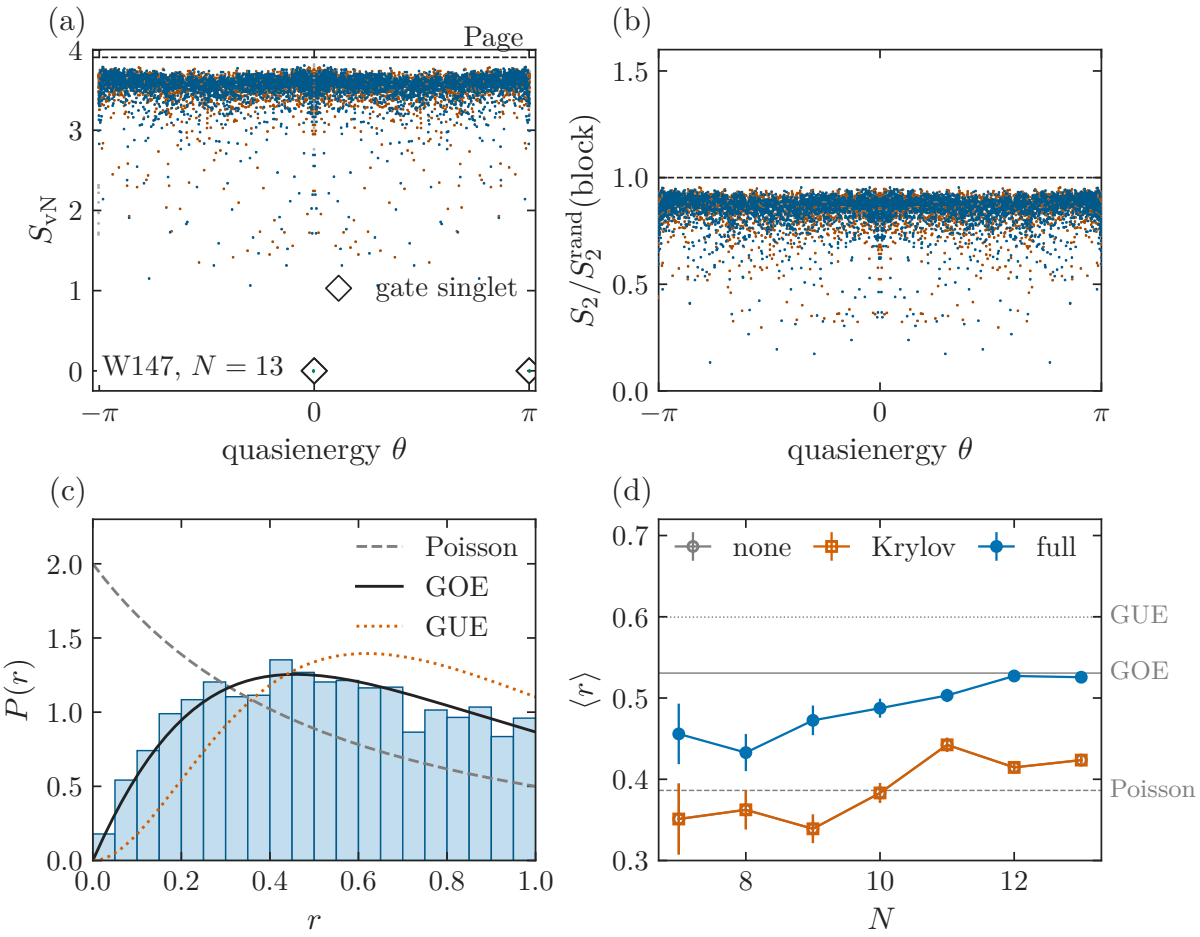


Figure 3: Rule 147 ($k = 3$), $N = 13$; panels as in Fig. 2. In (d) the Krylov-only statistics (here the whole spectrum) stay near Poisson, while resolving the spatial symmetry (R for odd N, the half step Ω for even N) brings $\langle r \rangle$ to the GOE value.

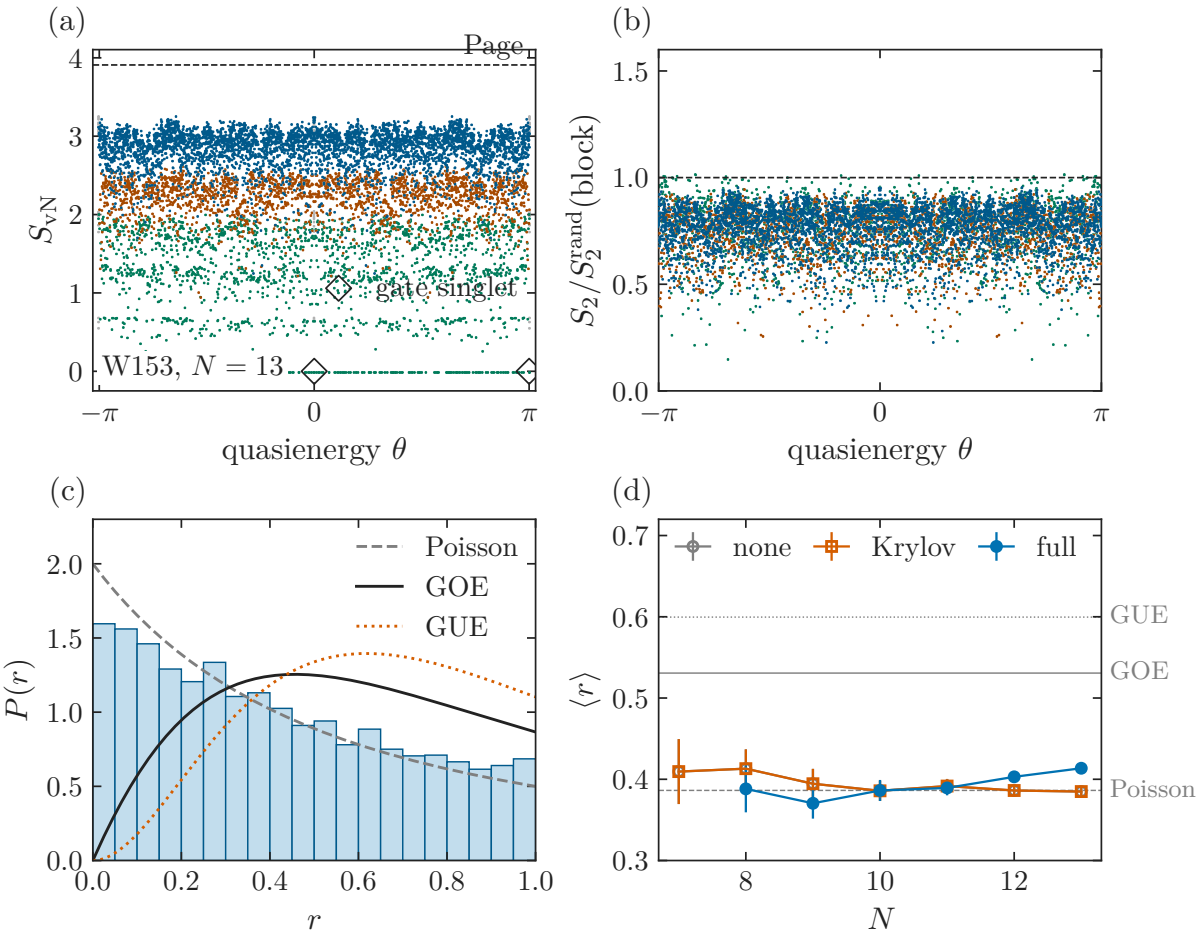


Figure 4: Rule 153 ($k = 2$, one-sided), $N = 13$; panels as in Fig. 2. The entanglement bands in (a) are the blocks of the boundary-driven tower (sizes $2^{N-1}, 2^{N-2}, \ldots$). $P(r)$ is close to Poisson, while $\langle r \rangle$ drifts upward with N.

sectors and 3 blocks, instead of $N + 1$ blocks), and 60 loses its many small blocks (3 blocks). The full comparison awaits the production runs.

6 The X gate

Not run (budget and the reaper stop). The expected structure is stated in Sec. 2: every sector is a single cycle, the spectrum is roots of unity, and $\langle r \rangle$ measures orbit arithmetic, not chaos.

7 Compute, memory, and how to resume

Compute used. Production started at 01:53 and was stopped at 02:51 (about 1 h of wall time; all runs on one 12th-gen i7, 31 GB). In that hour it produced 137 obc0 units: all 16 rules for $N = 6 \ldots 13$, plus $N = 14$ for 9 rules (the largest, rule 60/102 with a 2^{13}-dimensional sector, took 100 s). Validation and memory tests added about 20 minutes. The 8-hour budget is therefore largely unused.

What stopped it. The session's memory reaper stopped all background jobs while rule 51 at $N = 14$ was being processed (a single 16384-dimensional sector). Instrumenting the phases at $N = 12$ showed that the memory went *not* into the dense linear algebra but into U itself. For the chaotic rules U is effectively dense (each column has up to 2^N entries), so storing it as a sparse matrix, and running Graphs.jl's component search over that adjacency, cost 1.1 GB at $N = 12$. That extrapolates to about 18 GB at $N = 14$.

The fix (v2, `generate_spectral_data_v2.jl`).

1. U is never formed: $Uv = L_B(L_A v)$ with the two sparse layers (a few MB at $N = 14$).
2. Krylov sectors come from a union-find over the supports of the *individual gates*. For $V = H$ this is the same partition as the components of $\mathrm{supp}(U)$ (R-T19 Prop. 4), and it is asserted against the old construction for every $N \leq 10$ run. For $V = X$, U is a permutation and is formed directly.
3. Symmetry checks are random-vector tests through the layers. Dense sector blocks are assembled column-chunk-wise. Inside each block, the products AV, SV and EV are computed per batch of at most 2048 columns.

Validation: v2 reproduces v1 block by block (sector size, block size, sorted quasienergies, S_2^{Haar}, degeneracies) for seven rules at $N = 9$. Its pbc spectra match the exact Python engine to $< 10^{-14}$. Non-degenerate eigenvectors agree to 10^{-15} in S_{vN}; degenerate ones are basis dependent by construction.

Measured (rule 147, $N = 13$, 8192-dimensional sector): peak resident memory 4.5 GB (v1: 7.4 GB), run time 124 s (v1: 224 s). Expected peak at $N = 14$: about 15–16 GB.

How to resume (not done without permission). From the repository root:

```
bash results_spectra/run_H.sh      # obc0: rest of N=14, N=15-18
bash results_spectra/run_H_pbc.sh  # pbc, even N=6..14
bash results_spectra/run_X.sh      # optional: X gate
```

(all three scripts now call v2; finished units are skipped.) Estimated wall time: about 1.5 h (obc0 remainder), 1–1.5 h (pbc), and under 1 h (X): well inside the remaining budget.

A Bibliography checks

Every entry below was verified on 2026-10-14 against the arXiv listing, the DOI or publisher page, or an institutional repository record. The journal references of Regnault–Nandkishore and Ponte *et al.* (PRL 114, 140401) were confirmed from the arXiv journal-ref field. Deutsch (1991) was confirmed through citing papers only (Rigol *et al.* 2008 and the reviews). Jordan (1875) is cited for the origin of the two-subspace decomposition; the modern statement is Halmos (1969).

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